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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
18/103,058	01/30/2023	Syuuhei SHIRO	94377-1667 08TMCT170902PA	5934

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EXAMINER	
JACKSON, DANIELLE MARIE	

ART UNIT	PAPER NUMBER
3657	

NOTIFICATION DATE	DELIVERY MODE
03/17/2025	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

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Office Action Summary	Application No. 18/103,058	Applicant(s) SHIRO et al.	
	Examiner DANIELLE M JACKSON	Art Unit 3657	AIA (FITF) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 1/30/2023.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL.** 2b) ☑ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1-11 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1-11 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☑ The drawing(s) filed on 1/30/2023 is/are: a)☑ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☐ All b)☐ Some** c)☐ None of the:

1.☐ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☑ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date _____.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

DETAILED ACTION

This is the first office action in reply to U.S. application 18/103,058. Claims 1-11 are pending.

Notice of Pre-AIA or AIA Status

The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Claim Rejections - 35 USC § 112

The following is a quotation of 35 U.S.C. 112(b):
(b) CONCLUSION.—The specification shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the inventor or a joint inventor regards as the invention.

The following is a quotation of 35 U.S.C. 112 (pre-AIA), second paragraph:
The specification shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the applicant regards as his invention.

Claims 1-11 rejected under 35 U.S.C. 112(b) or 35 U.S.C. 112 (pre-AIA), second paragraph, as being indefinite for failing to particularly point out and distinctly claim the subject matter which the inventor or a joint inventor (or for applications subject to pre-AIA 35 U.S.C. 112, the applicant), regards as the invention.

Claims 1 and 10-11 cite the limitation "the regulated region being a region to which entry of a mobile body is regulated". The use of the phase "to which" makes it unclear what the elements of the claim limitation "which" is referring to. Examiner recommends amending this to "the regulated region being a region ~~to which~~ wherein entry of a mobile body is regulated". Claims 2-10 are rejected due to their dependency on claim 1.

Claim Rejections - 35 USC § 103

In the event the determination of the status of the application as subject to AIA 35 U.S.C. 102 and 103 (or as subject to pre-AIA 35 U.S.C. 102 and 103) is incorrect, any correction of the statutory basis (i.e., changing from AIA to pre-AIA) for the rejection will not be considered a new ground of rejection if the prior art relied upon, and the rationale supporting the rejection, would be the same under either status.

The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

Claims 1-6 and 8-11 are rejected under 35 U.S.C. 103 as being unpatentable over Suvarna (US 20190061157) in view of Shin (US 20170205822).

Regarding claim 1, Suvarna teaches *a regulated region management system comprising one or more processors* ([0027] discusses the system being operable with a processor where [0064] discusses the regulated region management) *configured to:*

acquire a hand-drawn input image input to an environment map ([0064] "the user uses a finger on a touch screen to draw a virtual boundary on a map")*;*

estimate a regulated region based on the environment map and the hand-drawn input image, the regulated region being a region to which entry of a mobile body is regulated ([0064]-[0065] discuss the virtual boundaries being used to determine a region designated by the user as either an area to be cleaned or avoided (regulated))*;*

transmit regulated region information that indicates the regulated region to the mobile body when the mobile body is not remotely manipulated using a manipulation terminal ([0076] discusses the user indicating the region using a user device and this being transmitted to the robot to operate autonomously using local coordinates)*;* **but does not explicitly teach**

transmit the regulated region information to the manipulation terminal when the mobile body is remotely manipulated using the manipulation terminal.

Shin teaches *transmit the regulated region information to the manipulation terminal when the mobile body is remotely manipulated using the manipulation terminal* ([0063] discusses a user input and a display unit for a user interface unit used to remotely control a mobile robot where the display unit displays a map and where [0081] discusses the user being able to input a designated area using a touch input (hand-drawn) with Figs. 6 and 7 showing the map including the designated area).

Suvarna teaches an autonomous mobile robot where a user is able to draw a virtual boundary using a remote device that is capable of remote control [0029]. Shin teaches a mobile robot that is capable of being remotely controlled by a user using in part displayed maps. It would have been obvious to a person having ordinary skill in the art before the effective filing date of the invention to modify the system of Suvarna with the remote control of Shin as Shin teaches that this improves the user's convenience [0029].

Regarding claim 2, Suvarna teaches *wherein the one or more processors are configured to transmit the regulated region information to the mobile body* ([0076]

discusses the user indicating the region using a user device and this being transmitted to the robot with Fig. 7 showing this occurs using the processing subsystem).

Regarding claim 3, Suvarna teaches *wherein the one or more processors are configured to:*

extract coordinate information and color information from the hand-drawn input image ([0064] discusses using colors as a visual indicator of the virtual boundary where [0076] further discusses the virtual boundary being converted into coordinates)*;*

estimate the regulated region based on the environment map and the coordinate information ([0076] discusses using the map and the coordinate information to provide the virtual boundaries to the robot)*;*

estimate a type of a regulation based on the color information ([0064] discusses using the color to visually indicate that an area should be cleaned or avoided)*; and*

generate information that indicates the regulated region and information that indicates the type of the regulation as the regulated region information ([0064] discusses determining the virtual boundary which includes the regulation based on color and [0076] discusses transmitting this information to the robot).

Regarding claim 4, Suvarna teaches *wherein the one or more processors are configured to:*

transmit, when identification information on the mobile body is acquired together with the hand-drawn input image and the mobile body is not remotely manipulated using the manipulation terminal, the regulated region information to the mobile body with the

identification information ([0076] discusses the user indicating the region using a user device and this being transmitted to the robot to operate autonomously using local coordinates where [0088]-[0089] discuss the use of multiple robots where each robot can be assigned different areas and tasks where it is interpreted that this would include identification information of the robot)*;* **but does not explicitly teach**

transmit, when the identification information on the mobile body is acquired together with the hand-drawn input image and the mobile body with the identification information is remotely manipulated, the regulated region information to the manipulation terminal that is used to remotely manipulate the mobile body with the identification information.

Shin teaches *transmit, when the identification information on the mobile body is acquired together with the hand-drawn input image and the mobile body with the identification information is remotely manipulated, the regulated region information to the manipulation terminal that is used to remotely manipulate the mobile body with the identification information* ([0063] discusses a user input and a display unit for a user interface unit used to remotely control a mobile robot where the display unit displays a map and where [0081] discusses the user being able to input a designated area using a touch input (hand-drawn) with Figs. 6 and 7 showing the map including the designated area and where [0057] discusses transmitting identification to a mobile robot to ensure connection between the robot and the user interface).

Suvarna teaches an autonomous mobile robot where a user is able to draw a virtual boundary using a remote device that is capable of remote control [0029]. Shin teaches a mobile robot that is capable of being remotely controlled by a user using in

part displayed maps. It would have been obvious to a person having ordinary skill in the art before the effective filing date of the invention to modify the system of Suvarna with the remote control of Shin as Shin teaches that this improves the user's convenience [0029].

Regarding claim 5, Suvarna teaches *a mobile body management system comprising: the regulated region management system according to claim 1* ([0064]-[0065] as described above in claim 1)*; and*

the mobile body ([0023]-[0024] discuss cleaning robot capable of moving).

Regarding claim 6, Suvarna teaches *wherein the mobile body is configured to generate a movement plan for autonomous movement of the mobile body based on the environment map and the regulated region information in response to receiving the regulated region information when the mobile body is autonomously moving* ([0076]-[0077] discuss the robot converting the global coordinates of the virtual boundaries into local coordinates which it can then use with its own localization to operate autonomously).

Regarding claim 8, Suvarna teaches a user operated device as an interface [0055] **but does not explicitly teach** *the manipulation terminal correlated with the mobile body and configured to display the regulated region in response to receiving the regulated region information, and transmit, when a manipulation instruction is received from a manipulator, the manipulation instruction to the corresponding mobile body.*

Shin teaches *the manipulation terminal correlated with the mobile body and configured to display the regulated region in response to receiving the regulated region information, and transmit, when a manipulation instruction is received from a manipulator, the manipulation instruction to the corresponding mobile body* ([0063] discusses a user input and a display unit for a user interface unit used to remotely control a mobile robot where the display unit displays a map and where [0081] discusses the user being able to input a designated area using a touch input (hand-drawn) with Figs. 6 and 7 showing the map including the designated area).

Suvarna teaches an autonomous mobile robot where a user is able to draw a virtual boundary using a remote device that is capable of remote control [0029]. Shin teaches a mobile robot that is capable of being remotely controlled by a user using in part displayed maps. It would have been obvious to a person having ordinary skill in the art before the effective filing date of the invention to modify the system of Suvarna with the remote control of Shin as Shin teaches that this improves the user's convenience [0029].

Regarding claim 9, Suvarna teaches *wherein the mobile body is configured to generate a movement plan to detour around the regulated region, or stop moving, in a case where a prediction is made that the mobile body enters the regulated region* ([0064] discusses the regulated region being a region that needs to be avoided) **but does not explicitly teach** *when the mobile body moves in accordance with the manipulation instruction received from the corresponding manipulation terminal.*

Shin teaches *when the mobile body moves in accordance with the manipulation instruction received from the corresponding manipulation terminal* (([0063] discusses a user input and a display unit for a user interface unit used to remotely control a mobile robot where the display unit displays a map and where [0081] discusses the user being able to input a designated area using a touch input (hand-drawn) with Figs. 6 and 7 showing the map including the designated area).

Suvarna teaches an autonomous mobile robot where a user is able to draw a virtual boundary using a remote device that is capable of remote control [0029]. Shin teaches a mobile robot that is capable of being remotely controlled by a user using in part displayed maps. It would have been obvious to a person having ordinary skill in the art before the effective filing date of the invention to modify the system of Suvarna with the remote control of Shin as Shin teaches that this improves the user's convenience [0029].

Regarding claim 10, Suvarna teaches *a regulated region management method* (Fig. 9) *comprising:*

acquiring a hand-drawn input image input to an environment map ([0064] "the user uses a finger on a touch screen to draw a virtual boundary on a map")*;*

estimating a regulated region based on the environment map and the hand-drawn input image, the regulated region being a region to which entry of a mobile body is regulated ([0064]-[0065] discuss the virtual boundaries being used to determine a region designated by the user as either an area to be cleaned or avoided (regulated))*;*

transmitting regulated region information that indicates the regulated region to the mobile body when the mobile body is not remotely manipulated using a manipulation terminal ([0076] discusses the user indicating the region using a user device and this being transmitted to the robot to operate autonomously using local coordinates)*;* **but does not explicitly teach**

transmitting the regulated region information to the manipulation terminal when the mobile body is remotely manipulated using the manipulation terminal.

Shin teaches *transmitting the regulated region information to the manipulation terminal when the mobile body is remotely manipulated using the manipulation terminal* ([0063] discusses a user input and a display unit for a user interface unit used to remotely control a mobile robot where the display unit displays a map and where [0081] discusses the user being able to input a designated area using a touch input (hand-drawn) with Figs. 6 and 7 showing the map including the designated area).

Suvarna teaches an autonomous mobile robot where a user is able to draw a virtual boundary using a remote device that is capable of remote control [0029]. Shin teaches a mobile robot that is capable of being remotely controlled by a user using in part displayed maps. It would have been obvious to a person having ordinary skill in the art before the effective filing date of the invention to modify the system of Suvarna with the remote control of Shin as Shin teaches that this improves the user's convenience [0029].

Regarding claim 11, Suvarna teaches *a non-transitory storage medium that stores instructions that are executable by one or more processors and that cause the one or more processors to execute functions* ([0038]-[0043] discuss the computing system architecture including processors and non-transitory storage) *comprising:*

acquiring a hand-drawn input image input to an environment map ([0064] "the user uses a finger on a touch screen to draw a virtual boundary on a map");

estimating a regulated region based on the environment map and the hand-drawn input image, the regulated region being a region to which entry of a mobile body is regulated ([0064]-[0065] discuss the virtual boundaries being used to determine a region designated by the user as either an area to be cleaned or avoided (regulated));

transmitting regulated region information that indicates the regulated region to the mobile body when the mobile body is not remotely manipulated using a manipulation terminal ([0076] discusses the user indicating the region using a user device and this being transmitted to the robot to operate autonomously using local coordinates); **but does not explicitly teach**

transmitting the regulated region information to the manipulation terminal when the mobile body is remotely manipulated using the manipulation terminal.

Shin teaches *transmitting the regulated region information to the manipulation terminal when the mobile body is remotely manipulated using the manipulation terminal* ([0063] discusses a user input and a display unit for a user interface unit used to remotely control a mobile robot where the display unit displays a map and where [0081] discusses the user being able to input a designated area using a touch input (hand-drawn) with Figs. 6 and 7 showing the map including the designated area).

Suvarna teaches an autonomous mobile robot where a user is able to draw a virtual boundary using a remote device that is capable of remote control [0029]. Shin teaches a mobile robot that is capable of being remotely controlled by a user using in part displayed maps. It would have been obvious to a person having ordinary skill in the art before the effective filing date of the invention to modify the system of Suvarna with the remote control of Shin as Shin teaches that this improves the user's convenience [0029].

Claim 7 is rejected under 35 U.S.C. 103 as being unpatentable over Suvarna in view of Shin and further in view of Patrick (US 20170341236).

Regarding claim 7, Suvarna teaches a movable robot but does not explicitly teach *wherein: the mobile body includes a movable portion that acts differently from movement of the mobile body in a travel direction; and the mobile body is configured to, when the mobile body is autonomously moving, estimate a movable range of the mobile body based on the regulated region information and the movement plan, and generate an operation plan for the movable portion of the mobile body based on the movable range.*

Patrick teaches *wherein: the mobile body includes a movable portion that acts differently from movement of the mobile body in a travel direction* ([0019] discusses a mobile robot with an attached manipulator arm where [0025] discusses the robotic vehicle and/or the manipulator moving based on detected information)*; and*

the mobile body is configured to, when the mobile body is autonomously moving, estimate a movable range of the mobile body based on the regulated region information

and the movement plan, and generate an operation plan for the movable portion of the mobile body based on the movable range ([0036]-[0037] discuss determining how to control both the robotic vehicle and the manipulator arm using a sequence of movements to perform the required tasks while avoiding collisions by determining how the robot moves (movable range)).

Suvarna teaches a robot that is capable of moving to avoid a user indicated region. Patrick teaches a movable robot with a manipulator arm that can avoid collisions. It would have been obvious to a person having ordinary skill in the art before the effective filing date of the invention to modify the system of Suvarna with the manipulator arm of Patrick as this would increase the functionality of the robot in performing necessary tasks improving the user's experience.

Conclusion

The prior art made of record and not relied upon is considered pertinent to applicant's disclosure: Chu (US 20180312033) teaches a user hand-drawing a safety zone; Ebrahimi Afrouzi (US 20220066456) teaches a user drawing lines in an app to split a map of a robot into sections; and Swann (US 20240246249) teaches collaborative movement between autonomous vehicle and an attached robotic manipulator.

Any inquiry concerning this communication or earlier communications from the examiner should be directed to DANIELLE M JACKSON whose telephone number is (303)297-4364. The examiner can normally be reached Monday-Friday 7:00-4:30 MT.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Abby Lin can be reached on (571) 270-3976. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

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/D.M.J./
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